

EXCLUSIVE

IAS PRELIMS 2026

CURRENT AFFAIRS

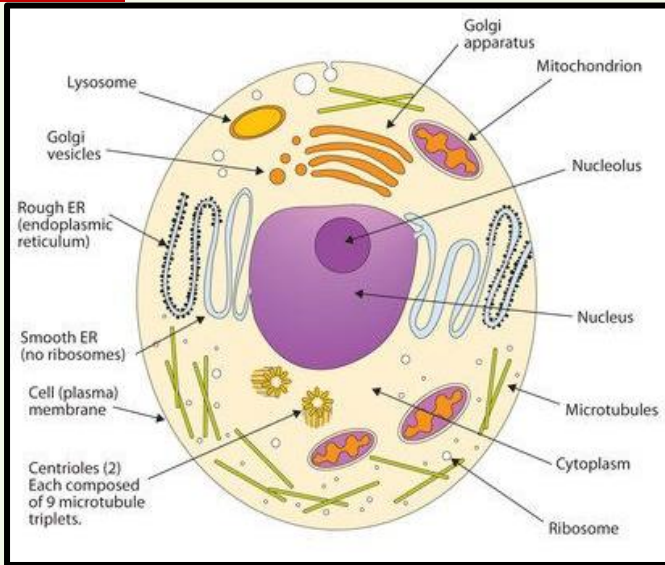
REVISION

GS-1 (DYNAMIC)

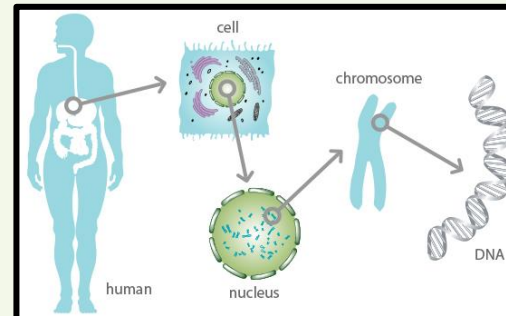
MODULE - 42

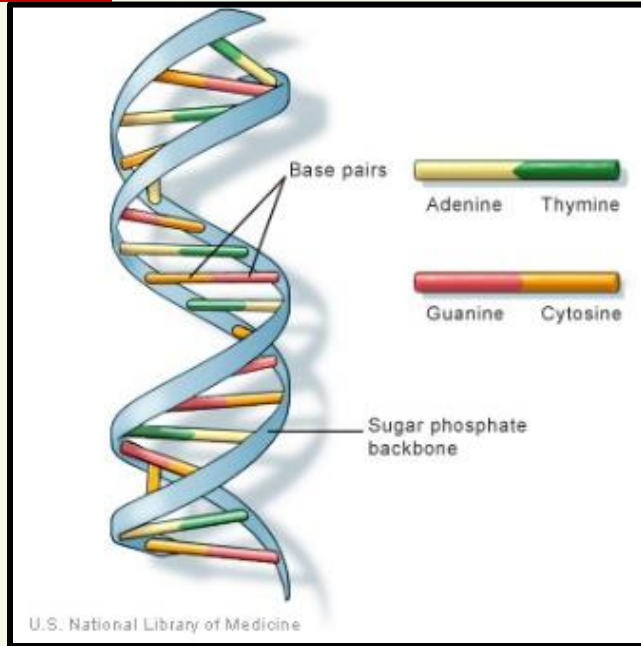


**SCIENCE
&
TECHNOLOGY-10**

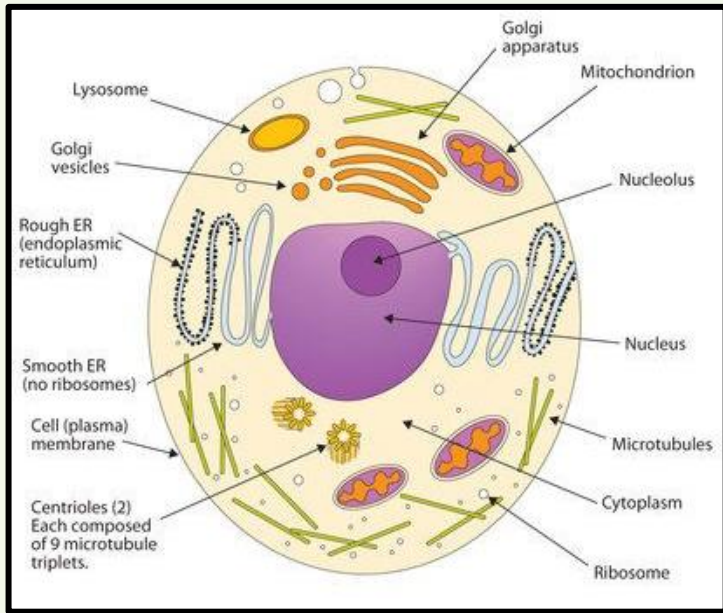


- Gene therapy is a technique that uses a gene(s) to treat, prevent or cure a disease or medical disorder. Often, it works by adding new copies of a gene that is broken, or by replacing a defective or missing gene in a patient's cells with a healthy version of that gene.
- Hence, it is permanently altering the hard copy of the genetic instructions in the DNA.
- It is a one-time treatment and is being used only for diseases that have no other cures.
- It can be used to treat or prevent diseases, which is possible by altering the genes in somatic / germline cells.
- It is considered highly risky, if not done properly.





- CRISPR stands for **Clustered Regularly Interspaced Short Palindromic Repeats**.
- Scientists say that it is **cost effective** editing of any genome by **guide RNA**, which recognizes the target sequence.
- This is very efficient method of making site specific **mutations** (changing the structure of a gene), contrary to the **random mutations** caused by conventional chemical or radiation technology.
- **Cas9** (CRISPR associated protein 9) is a protein which is used to **cut DNA**.
- In nature, bacteria uses the CRISPR system to protect themselves from **invading viruses**.
- CRISPR can be deployed in all types of organisms, such as plant and animal species, as well as humans.
- At present, India does not have any legally enforceable regulations in place.



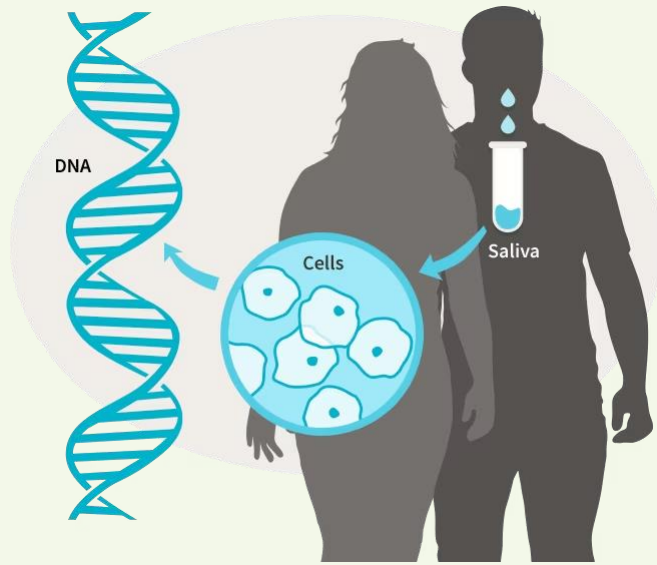
- Gene is part of our DNA that contains blueprint for making proteins such as hormones, enzymes etc.
- Medicines kill the messenger in a process called RNA Interference.
- Original DNA gets untouched, but it targets the instructions. Tweaking the levels of protein will affect the course of disease.
- The treatment is continuous for better results.





- In the human body, most of the DNA in a genome is packed inside cells, whereas some fragments of Nucleic Acids are present outside the cell, in body fluids, which are known as cell-free DNA (cfDNA).
- cfDNA can be generated and released from a cell in a number of possible situations, including when a cell is dying, and the nucleic acids become degraded. Hence, the amount, size, and source of the cfDNA can vary across a range as well.
- One of the most widely used applications of cfDNA has been in screening fetuses for specific chromosomal abnormalities, an application known as non-invasive pre-natal testing.
- Researchers around the world are increasingly finding cfDNA to be a useful tool to understand human diseases and to use the knowledge to improve diagnosis, monitoring, and prognosis.





- Ancestry DNA (aDNA) studies powered by cutting edge genomic techniques provides unprecedented insights into the genetic makeup of our ancestors.
- By extracting and analyzing DNA from ancient skeletal remains, scientists can reconstruct the genetic profiles of these people.
- Studies of such ancestry DNA have provided glimpses into the genetic diversity and population dynamics of ancient communities, their migration patterns, interactions and adaptations to local environments and even into the diseases these people confronted and how the affiliations shaped human evolution.

What do you understand by the terms “bio-hacking” and “cybernetic organism”?



- Bio-hacking is the biological experimentation done to improve the qualities or capabilities of living organisms. In a nutshell, it is a body modification with the aim of gaining enhanced abilities. It can be by the use of drugs / implants / gene editing.
- It works outside a traditional medical or scientific research environment. It is a niche field. Already chips are being implanted under the skin and used as remote keys, wallets and data storage devices.
- Cybernetic Organism, also known as ‘Cyborg’, applies to an organism that has restored functions or enhanced abilities due to the integration of some artificial component or technology.



- It is the content that is not indexed by Search Engines.
- It requires special/specific software, configuration, authorization to access.
- It is an overlay network, i.e., computer network that is layered on another network.

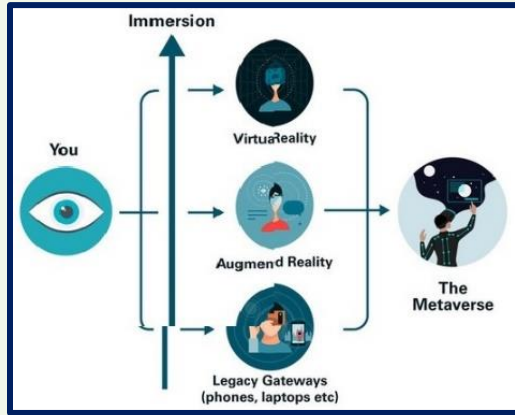


What do you understand by the terms “Visible Universe”, “Dark Matter”, “Dark Energy”, “Blue Sky Research”?



- “Visible Universe” constitutes visible or baryonic matter, which constitutes Earth, Sun, Stars, Galaxies and all this constitutes around 4.6%, say roughly 5%.
- Rest of the Universe appears to be made of mysterious invisible substance and it is “Dark Matter”, which constitutes around 27%. “Dark Energy” ripples gravity and causes expansion of the Universe and it is around 68%.
- “Blue Sky Research” is the research in fields like quantum computing, neural networks, dark matter, dark energy, genomics, etc.





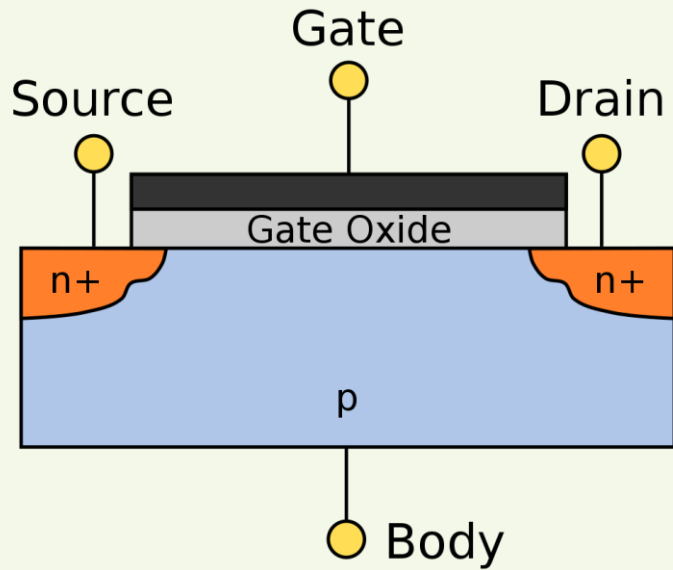
- Metaverse was first coined by science fiction writer Neil Stephenson in 1992 for his Novel, Snow Crash.
- It is a virtual environment that one can go inside of, instead of looking at something on the screen.
- The gadgets used are virtual reality headsets, smartphone apps, and augmented reality glasses.
- It is a three-dimensional version of Internet, and each one of us will have an Avatar and interaction with other Avatars take place and this amalgamates the elements of virtual reality and augmented reality.
- Hence, it is interconnected 3D Virtual World by using Augmented Reality (AR), Virtual Reality (VR), Artificial Intelligence (AI) and Blockchain.
- It gives immersive experience and makes gaming more realistic and enhances the user experience.

One could attend a virtual concert, try on digital clothing and go on trip with friends and visit the zoo. On the laptop screen, one can meet and interact with each other in a virtual house.

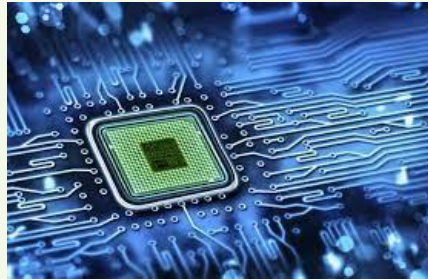
Rubella is a mild viral infection but has serious consequences, if infection occurs in pregnant women. It causes Congenital Rubella Syndrome (CRS). Newborn babies will be affected with cataract, glaucoma, hearing loss, mental retardation, microcephaly as well as heart defects.



- Measles is an acute viral respiratory illness.
- Humans are the only natural hosts of measles virus.
- It is one of the most contagious of all infectious diseases and up to 9 out of 10 susceptible persons with close contact to a measles patient will develop measles. Measles virus can remain infectious in the air for up to two hours after an infected person leaves an area.
- Measles viruses kill immune cells, leaving the child vulnerable to infectious diseases for 2 to 3 years.
- Measles can be prevented with measles-containing vaccine, which is primarily administered as the combination Measles-Mumps-Rubella (MMR) vaccine.
- Measles vaccine not only provides lifelong protection against virus, but also reduces mortality from other childhood infections.
- The measles vaccine was introduced in India's Universal Immunization Programme in 1985 and to further boost the programme, a second dose was introduced in 2010.
- To control measles and prevent outbreaks and deaths, vaccination coverage rates must reach 95% and be maintained at national and sub-national levels.

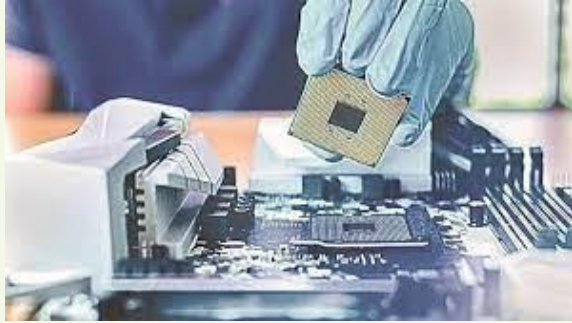


- Transistor has three parts – the source, gate and the drain. The flow of current between the source and the drain (sink) is regulated by the voltage applied to the gate.
- By manipulating the gate to open or close, the transistors stores and manipulates the data in a semiconductor chip. The semiconductor stores information in the form of bits.
- If there are more bits, a semiconductor can store and more quickly it can manipulate them i.e., the more data transistors can process.

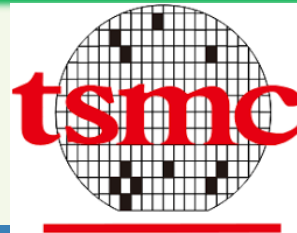


Each square inch of a chip contains billions of transistors. The distance between two transistors is measured in nanometers.

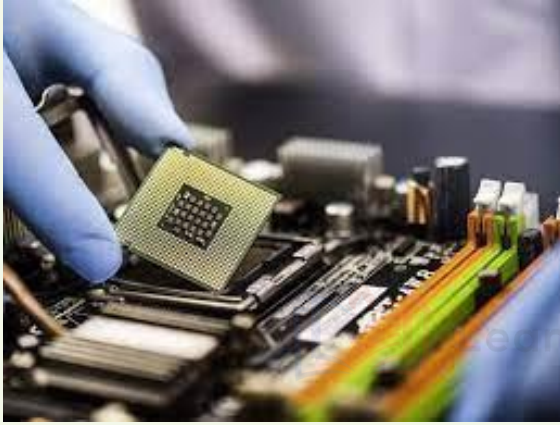
What do you know about the Global Value Chains in the Semiconductor Ecosystem?



- The USA leads the World by controlling intellectual property, design and technology.
- Japan makes silicon wafers on which chip circuitry is etched.
- For etching this circuitry onto the wafers, Netherlands leads in producing lithography machines.
- The fabrication units, which manufacture chips are concentrated in Taiwan, China and South Korea. Taiwanese company TSMC leads the world in the manufacturing of advanced chips - 5nm and 3nm. Now research is going on to develop 2nm chips.
- Each square inch of a chip contains billions of transistors. The distance between two transistors is measured in nanometers.



What are the challenges involved in the Chip Manufacturing through Fabs?



- Chip fabs require millions of liters of clean water and extremely stable power supply.
- It costs multiples of a billion dollars to set up fab.
- In short span of time, manufacturing facilities may become outdated.
- To stay ahead of the game, semiconductor firms invest about quarter of their annual revenue in R&D, the highest for any industry.

India is rightly concentrating on medium chips of 28 nm, as these power white goods, low end phones, medical devices, other electronic goods, etc.



- Urea Gold fertilizer is developed by Rashtriya Chemicals and Fertilizers (RCF) Ltd.
- It has 37% Nitrogen and 17% Sulphur. Normal Urea contains 46% Nitrogen.
- Urea Gold fulfills Sulphur requirements in Indian soils and enhances nitrogen use efficiency. Sulphur is particularly crucial for oil seeds and pulses.
- The cost of 45 kg of normal neem coated urea is Rs. 266.50, whereas 40 kg urea gold costs the same.





- Cruise missiles are jet-propelled at subsonic speeds throughout their flight, while ballistic missiles are rocket powered only in the initial phase of flight, after which they follow an arcing trajectory to the target.
- Intercontinental ballistic missiles have a range beyond 5,500 kilometers.
- Cruise missiles fly in a relatively straight line and at lower altitudes. We can say cruise missiles are unmanned vehicles, that are propelled by jet engines, much like an airplane. They remain within the atmosphere for the duration of their flight.
- Cruise missiles are self-guided and use multiple methods to accurately deliver their payload, including terrain mapping, global positioning systems etc.

What do you know about the first Liquefied Natural Gas (LNG) powered Bus?



- The Chief Minister of Maharashtra inaugurated India's first LNG powered bus.
- They can travel long distances in one filling. They have a driving range of 600 to 1,000 kilometers in a single fill giving it an edge versus equivalent CNG buses.
- LNG vehicles have 30% lower CO₂ Emissions, 90% lower NO_x and PM emissions and 100% lower SO_x emissions versus diesel vehicles.
- They are theft-free due to their cryogenic nature.
- Existing diesel vehicles can be retrofitted to use LNG.

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For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

For Information:

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